

Lab 6: Exceptions and sorting

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Space Station Management

We will incorporate exceptions into the Space Station Management project.

0.1 Data validation

Use *IllegalArgumentException* in constructors and setters where needed. For example:

- crew member age and mission hours cannot be negative,
- officer years of experience cannot be negative,
- supply quantity and cost cannot be negative,
- module capacity must be greater than zero,
- access card dates must be valid.

This validation should also be considered in other classes.

0.2 CrewMember

Use *Comparable* and *Comparator* interfaces to sort crew members by:

- age ascending,
- role descending and mission hours ascending.

Add a boolean variable *isConsumable* to class **Supply**, and a method *useSupply(Supply supply)* to the class representing a crew member.

Implement an *InvalidSupplyException* class to handle cases where a crew member is given a supply that should not be consumed.

0.3 Module

Implement a *ModuleFullException* that is thrown when attempting to add too many crew members to a module.

0.4 AccessCard

Implement an *InvalidAccessCardDateException* class to handle cases where:

- an access card is created with an issue date from the past,
- a premium access card has an expiration date earlier than its issue date.

0.5 Tester

In the tester class, create a list of **CrewMember** objects, including instances of child classes such as **Engineer**, **Scientist**, **Pilot**, and **Medic**. Then sort the list and display the results.

Use **Supply** objects and test the *useSupply()* method for different crew members. Add a method in **Supply** that returns the total price, assuming that *cost* is the price of a single unit and *quantity* is the number of units. Calculate and summarize the total supply expenses.

Also test:

- adding too many crew members to a module,
- creating invalid access cards,
- assigning invalid supplies to crew members.

Make sure that exception handling is properly tested with *try-catch* blocks.